

STAT/MA 41600

In-Class Problem Set #5: September 4, 2026

1. In a recent study that focuses only on the top 3 Generative AI assistants, it is estimated that 50% of users prefer ChatGPT, 33% prefer Gemini, and 17% prefer Claude. (For this question, assume that people's preferences are independent.)

Suppose that ChatGPT is correct with probability $8/10$; Gemini is correct with probability $9/10$; and Claude is correct with probability $13/15$.

1a. If you meet a person who used their preferred Generative AI, and the AI was correct, what is the probability that they are using ChatGPT?

1b. If you meet a person who used their preferred Generative AI, and the AI was correct, what is the probability that they are using Gemini?

1c. If you meet a person who used their preferred Generative AI, and the AI was correct, what is the probability that they are using Claude?

1d. If you pick a person at random, what is the probability that, when they use their preferred Generative AI model, the AI will be incorrect?

2. Suppose that Bob rolls two 6-sided dice, and the sum of the dice is 7 or greater. What is the conditional probability that the sum of the dice is exactly 7?

3. Suppose that Alice draws two cards from a standard 52-card deck, **without replacement**. What is the probability that the values on the two cards agree? (Do not pay attention to the suits; only pay attention to the values.)

4. Alice rolls a 6-sided die. Whatever value she gets, Bob tosses that many fair coins. (For example, if Alice rolls a "5", then Bob tosses 5 fair coins.)

4a. What is the probability that Bob has 0 heads?

4b. What is the probability that Bob has 1 head?

4c. What is the probability that Bob has 2 heads?

4d. What is the probability that Bob has 3 heads?

4e. What is the probability that Bob has 4 heads?

4f. What is the probability that Bob has 5 heads?

4g. What is the probability that Bob has 6 heads?

5. Alice rolls a 6-sided die. Whatever value she gets, Bob tosses that many fair coins. (For example, if Alice rolls a "5", then Bob tosses 5 fair coins.)

Suppose that Bob has 0 heads (i.e., all of his flips are tails).

5a. What is the conditional probability that Alice rolled a value of 1?

5b. What is the conditional probability that Alice rolled a value of 2?

5c. What is the conditional probability that Alice rolled a value of 3?

5d. What is the conditional probability that Alice rolled a value of 4?

5e. What is the conditional probability that Alice rolled a value of 5?

5f. What is the conditional probability that Alice rolled a value of 6?